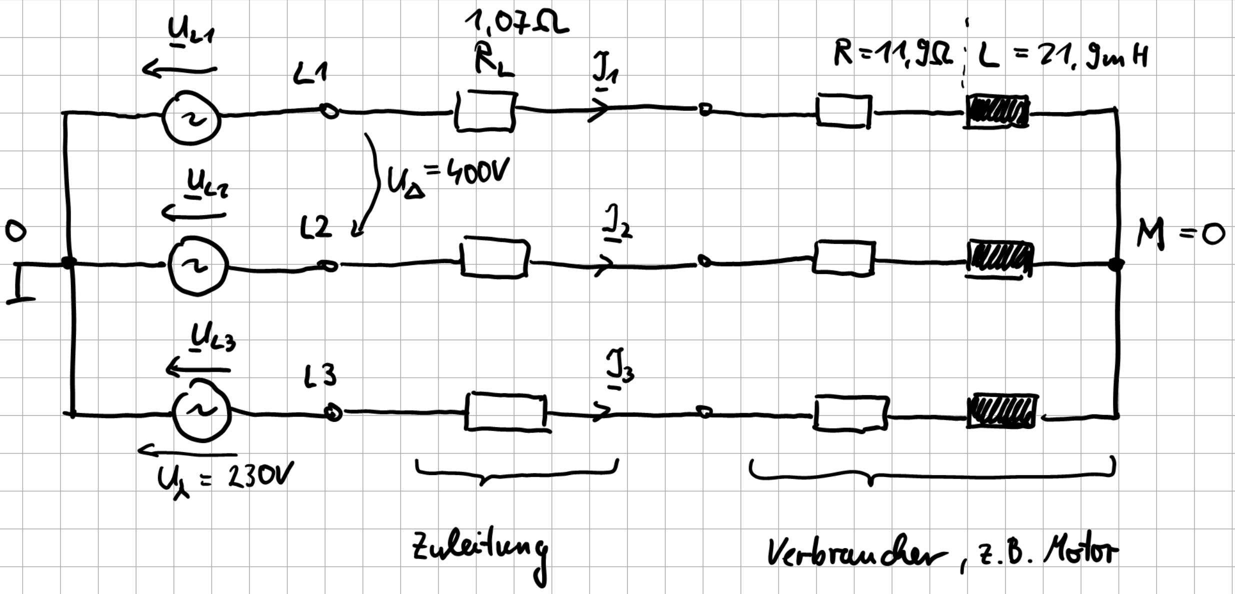
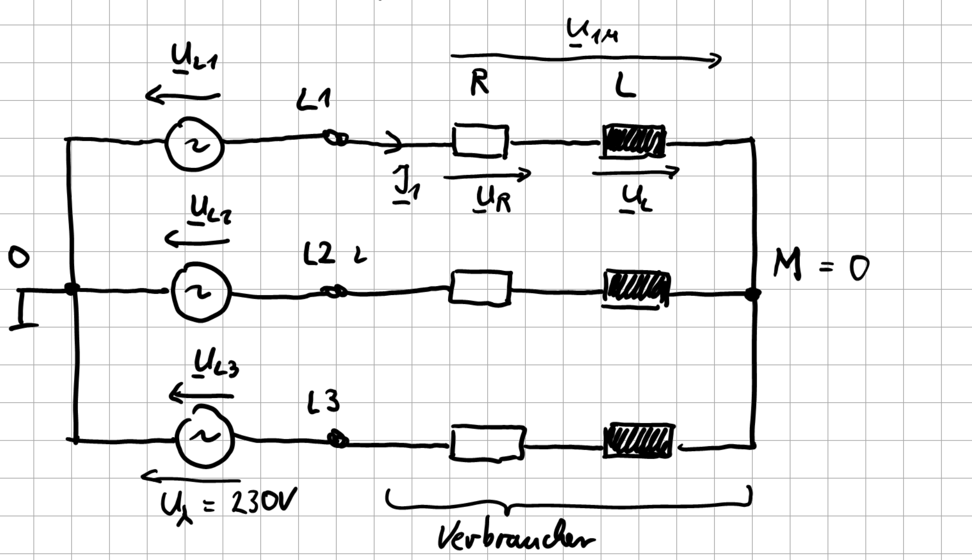
Übungsaufgabe symDreh5



$\cos\varphi = 0{,}866$

a) Infos vom Verbraucher: bei $U_\Delta = 400V$ ist $P = 10kW$

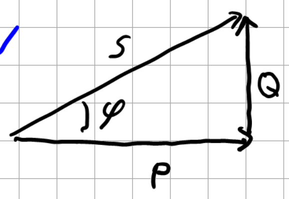
$\cos\varphi = 0{,}866$, ($f = 50\,Hz$)

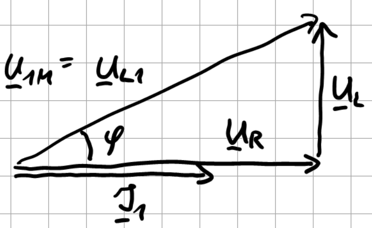


$P = 10kW$, $\cos\varphi = 0{,}866$ induktiv

$$\boxed{\begin{aligned} P &= 3\cdot U_\lambda \cdot I_\lambda \cdot \cos\varphi \\ Q &= 3\cdot U_\lambda \cdot I_\lambda \cdot \sin\varphi \\ S &= 3\cdot U_\lambda \cdot I_\lambda \end{aligned}}$$

$= 10kW$





$$\underline{U}_{1M} = R \cdot \underline{I}_1 + j\omega L \cdot \underline{I}_1$$

Wirkleistung nur am ohmschen Widerstand

$$P_R = U_R \cdot I_R = U_R \cdot I_1 = R \cdot I_1 \cdot I_1 = R \cdot I_1^2 = \frac{P_{ges}}{3} = 3{,}33\,kW$$

$$U_R = U_{1M} \cdot \cos\varphi = 230V \cdot 0{,}866 = 199{,}2V$$

$$U_L = U_{1M} \cdot \sin\varphi = 230V \cdot \sin 30° = 115\,V$$

$$\varphi = \arccos 0{,}866 = 30°$$

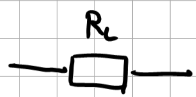
$$P_R = \frac{U_R^2}{R} \quad \Rightarrow \quad R = \frac{U_R^2}{P} = \frac{(199{,}2V)^2}{3{,}33\,kW} = 11{,}9\Omega$$

$$I_R = I_1 = \frac{U_R}{R} = \frac{199{,}2V}{11{,}9\Omega} = 16{,}7A$$

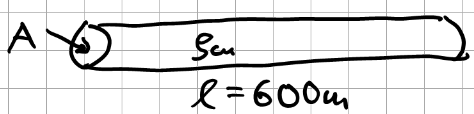
$$U_L = \omega L \cdot I_1$$

$$\Rightarrow L = \frac{U_L}{\omega \cdot I_1} = \frac{115V}{2\pi \cdot 50\frac{1}{s} \cdot 16{,}7A} = 21{,}9\,mH$$

b) R_L (Leitung)



$l = 600m$, $A = 10\,mm^2$, $\rho_{cu} = 0{,}01785\,\frac{\Omega mm^2}{m}$

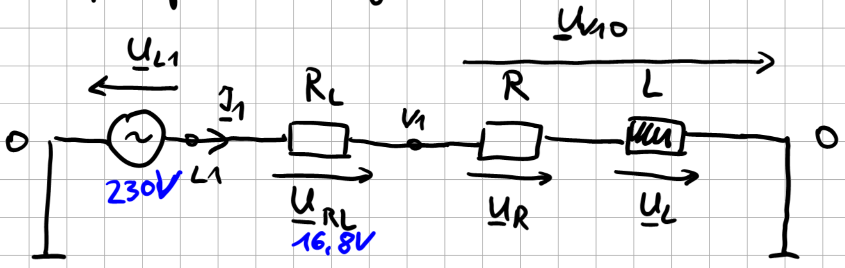


$$R_L = \frac{l}{A} \cdot \rho_{cu}$$

$$= \frac{600m}{10\,mm^2} \cdot 0{,}01785\,\frac{\Omega mm^2}{m}$$

$$= 1{,}071\Omega$$

c) einphasige Rechnung $\quad |\underline{U}_{V10}| = 215{,}8\,V$



$$\underline{U}_{RL} + \underline{U}_R + \underline{U}_L = \underline{U}_{L1}$$

$$R_L \cdot \underline{I}_1 + R \cdot \underline{I}_1 + j\omega L \cdot \underline{I}_1 = \underline{U}_{L1}$$

$$(R_L + R)\,\underline{I}_1 + j\omega L \cdot \underline{I}_1 = \underline{U}_{L1}$$

$$\big((R_L + R) + j\omega L\big)\,\underline{I}_1 = \underline{U}_{L1}$$

$$\underline{I}_1 = \frac{\underline{U}_{L1}}{R_L + R + j\omega L}$$

$$I_1 = \frac{U_{L1}}{\sqrt{(R_L+R)^2 + (\omega L)^2}}$$

$$= \frac{230V}{\sqrt{(11{,}9 + 1{,}07)^2\Omega^2 + (2\pi\, 50\tfrac{1}{s} \cdot 21{,}9mH)^2}}$$

$$= 15{,}7A$$

$$\underline{U}_{RL} = R_L \cdot \underline{I}_1$$

$$U_{RL} = R_L \cdot I_1 = 1{,}07\Omega \cdot 15{,}7A = 16{,}8V$$

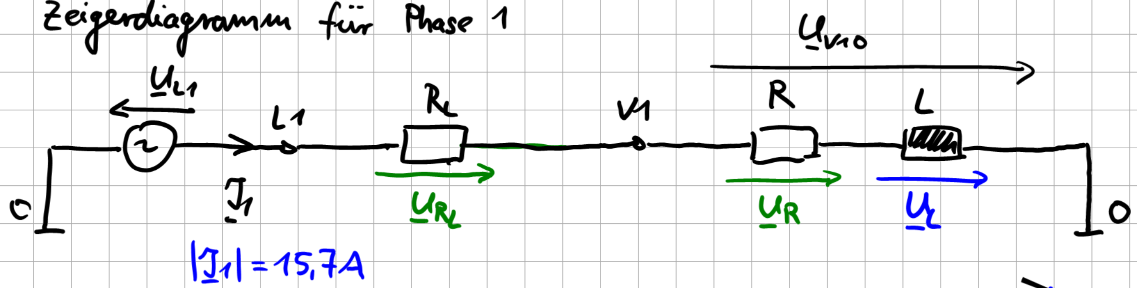
$$\underline{U}_{V10} = (R + j\omega L)\,\underline{I}_1$$

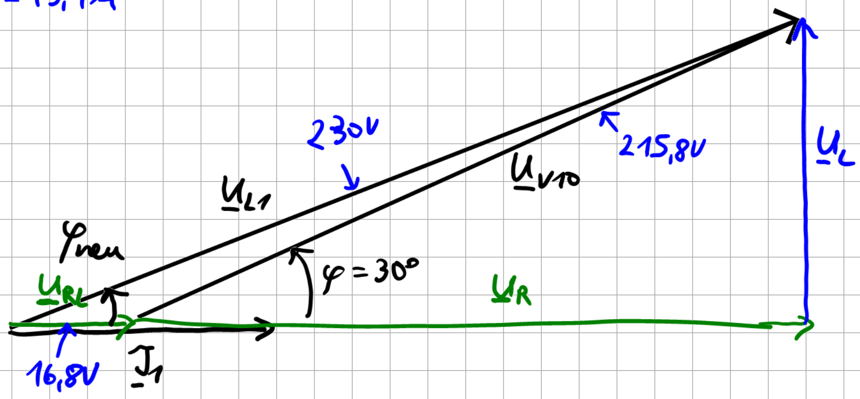
$$U_{V10} = \sqrt{R^2 + (\omega L)^2} \cdot I_1$$

$$= \sqrt{(11{,}9\Omega)^2 + (2\pi\, 50\tfrac{1}{s} \cdot 21{,}9mH)^2} \cdot 15{,}7A$$

$$= 215{,}8V$$

Zeigerdiagramm für Phase 1





d) Leitungsverluste

$$P_{R_L} = 3 \cdot U_{RL} \cdot I_1 = 3 \cdot 16{,}8V \cdot 15{,}7A$$

$$= 791 W$$

Wirkleistung Last

$$P_R = 3 \cdot R \cdot I_1^2 = 3 \cdot 11{,}9\Omega \cdot (15{,}7A)^2$$

$$= 8800 W$$

$$\frac{P_{RL}}{P_R} = \frac{791 W}{8800 W} = 9\%$$

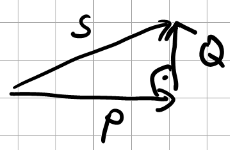
e) Verbraucherleistungen:

$P_{Verbraucher} = P_R = 8{,}8 kW$ ⇒ Differenz für Fall ohne Leitung 1,2 kW

$$S_{Verbraucher} = U_{1V0} \cdot I_1 \cdot 3$$

$$= (215{,}8V \cdot 15{,}7A) \cdot 3$$

$$= 10164 VA$$



$$Q_{Verbraucher} = \sqrt{(8{,}8kW)^2 + (10{,}16kVA)^2}$$

$$= 5{,}08 kvar$$

f) Netzleistungen:

$$S_{Netz} = 3 \cdot U_{L1} \cdot I_1$$
$$= 3 \cdot 230V \cdot 15{,}7A$$
$$= 10{,}83\ kVA$$

$$P_{Netz} = P_{Leitung} + P_{Verbraucher}$$
$$= 791W + 8800W$$
$$= 9591W$$

$$Q_{Netz} = \sqrt{S_{Netz}^2 - P_{Netz}^2}$$
$$= 5{,}08\ kvar$$